

Linear Algebra

A proof set for the Leibniz proof-verifier

This document presents five theorems from undergraduate linear algebra, each accompanied by an informal argument and a formal Lean 4 proof. Upload this PDF to the Leibniz engine to verify every theorem against the three gates: Validity, Alignment, and Reading.

Throughout, we model vectors as functions $\text{Fin } n \rightarrow \mathbb{R}$ (pointwise operations). This reduces vector-space axioms to the corresponding field axioms of the real numbers, applied componentwise via the `ext` tactic.

THEOREM 1

1. Commutativity of vector addition

Statement: For all vectors v, w in $\mathbb{R}^n$, $v + w = w + v$.

Informal argument. Vector addition is defined pointwise: $(v + w)(i) = v(i) + w(i)$. Since addition of real numbers is commutative, $v(i) + w(i) = w(i) + v(i)$ for every component i . The extensionality tactic `ext i` reduces the goal to equality at each component, where `add_comm` closes it.

FORMAL LEAN 4 PROOF

```
theorem add_comm_vec (v w : Fin n -> R) : v + w = w + v
  by ext i; exact add_comm (v i) (w i)
```

THEOREM 2

2. Associativity of vector addition

Statement: For all u, v, w in $\mathbb{R}^n$, $(u + v) + w = u + (v + w)$.

Informal argument. Again pointwise: $((u + v) + w)(i) = (u(i) + v(i)) + w(i)$, which equals $u(i) + (v(i) + w(i))$ by associativity of real addition. The same `ext i` / `add_assoc` pattern discharges the goal.

FORMAL LEAN 4 PROOF

```
theorem add_assoc_vec (u v w : Fin n -> R) : (u + v) + w = u + (v + w)
  by ext i; exact add_assoc (u i) (v i) (w i)
```

THEOREM 3

3. Distributivity of scalar multiplication over addition

Statement: For all c in R and v, w in R^n , $c * (v + w) = c * v + c * w$.

Informal argument. Scalar multiplication acts pointwise: $(c * (v + w))(i) = c * (v(i) + w(i))$. By distributivity of multiplication over addition in R , this equals $c * v(i) + c * w(i) = (c * v + c * w)(i)$. The tactic `mul_add` supplies the field law at each component after `ext i` opens the extensionality goal.

Remark. This is the first non-trivial vector-space axiom: it couples the additive and multiplicative structure of the field. Getting it right componentwise is the core check that R^n is genuinely an R -module, not merely an abelian group.

FORMAL LEAN 4 PROOF

```
theorem smul_add_vec (c : R) (v w : Fin n -> R) : c * (v + w) = c * v + c * w
  by ext i; exact mul_add c (v i) (w i)
```

THEOREM 4

4. Left distributivity over scalar addition

Statement: For all c, d in R and v in R^n , $(c + d) * v = c * v + d * v$.

Informal argument. Pointwise: $((c + d) * v)(i) = (c + d) * v(i) = c * v(i) + d * v(i)$ by left distributivity in R . The `add_mul` lemma discharges each component.

FORMAL LEAN 4 PROOF

```
theorem add_smul_vec (c d : R) (v : Fin n -> R) : (c + d) * v = c * v + d * v
  by ext i; exact add_mul c d (v i)
```

THEOREM 5

5. The identity map has eigenvalue one

Statement: For every nonzero vector v in R^n , the identity transformation satisfies $\text{id}(v) = 1 * v$.

Informal argument. The identity map leaves every vector unchanged: $\text{id}(v) = v$. By the unity law of scalar multiplication, $1 * v = v$. Hence $\text{id}(v) = 1 * v$, so every nonzero vector is an eigenvector of the identity with eigenvalue 1. This is the trivial case of the spectral theorem: the identity is already diagonal.

Note on the nonzero hypothesis. Eigenvectors are conventionally required to be nonzero so that the eigenvalue is uniquely determined. If $v = 0$ then $\text{id}(v) = \lambda * v$ for every λ , which would make the eigenvalue ill-defined.

FORMAL LEAN 4 PROOF

```
theorem eigenvalue_id (v : Fin n -> R) (hv : v != 0) : (fun x : Fin n -> R => x) v = (1 : R)
  * v
  by simp
```

THEOREM 6

6. Unity law for scalar multiplication

Statement: For every vector v in R^n , $1 * v = v$.

Informal argument. Multiplying each component by one leaves it unchanged: $(1 * v)(i) = 1 * v(i) = v(i)$. This is the multiplicative identity law of R lifted pointwise. The `simp` tactic closes it using the core `simp` lemma `one_mul`.

FORMAL LEAN 4 PROOF

```
theorem one_smul_vec (v : Fin n -> R) : (1 : R) * v = v
  by ext i; simp
```

End of proof set. Upload this PDF at leibniz.streamlit.app to verify all six theorems against the three gates.